

- 1 *Batrachochytrium dendrobatidis* is a fungus that lives in a warm and damp environment, with optimal temperatures for growth of between 17°C and 25°C.

B. dendrobatidis causes a disease in the skin of frogs, which can be fatal in up to 100% of infected frogs. The disease is known as chytridiomycosis.

The mountain chicken frog, *Leptodactylus fallax*, is a large species of frog and is critically endangered. The frog is native to the Caribbean islands of Dominica and Montserrat.

Populations of *L. fallax* were believed to be in some decline in the years before 1995, but between 1995 and 2011 the total number of frogs decreased by more than 88%.

- In 1995, ash and acid rain from a volcanic eruption killed a large number of frogs on Montserrat.
- In 2002, chytridiomycosis arrived on Dominica where it caused a severe decrease in the number of frogs within 18 months.
- In 2009, chytridiomycosis arrived on Montserrat where it caused a severe decrease in the number of frogs within 12 months.

Figure 1.1 shows a mountain chicken frog.



Figure 1.1

- (a) Suggest reasons that may have contributed to *L. fallax* becoming critically endangered, **other than** the volcanic eruption and the effects of chytridiomycosis.

.....

.....

.....

.....

.....

.....

.....

..... [3]



- (b) Captive breeding was used to increase the number of *L. fallax* individuals. Releasing these frogs into the wild was **not** successful because *B. dendrobatidis* was still present on Dominica and Montserrat.

Conservationists have trialled the re-introduction of *L. fallax* into safer, semi-wild areas on Dominica and Montserrat. Here, conditions have been created that protect *L. fallax* from infection by *B. dendrobatidis* but still allow the frog to survive and reproduce.

Suggest measures that could have been taken in these areas to protect *L. fallax* from chytridiomycosis.

.....

.....

.....

.....

.....

.....

..... [3]

- (c) Outline the characteristic features of the kingdom Fungi.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 10]